Vijay G K

Software Developer

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Summary

Software Developer with 4+ years of experience specializing in Python automation, full-stack web development, and enterprise RPA solutions within global banking environments. Adept at designing and deploying intelligent automation tools that reduce manual efforts while enhancing operational efficiency, and align with audit and compliance standards. Proven expertise in leveraging frameworks such as Django, ReactJS, and Selenium to build scalable systems across critical business functions. Recognized for driving end-to-end project delivery—from solution design to deployment—while collaborating with cross-functional teams across Barclays and Credit Suisse.

Work experience

Barclays - Automation Engineer 2023 - Present

Barclays Investment Bank is a global financial institution providing a comprehensive range of investment banking services, including advisory, financing, and risk management solutions to corporations, governments, and institutions worldwide.

Project Title: CAPTURE – Committee Report Consolidation Platform

Description: CAPTURE is a web-based application developed to streamline the committee report consolidation process within the business. Previously, approvals for individual PowerPoint slides were managed manually via team emails, resulting in version control issues, fragmented communication, and inefficiencies in consolidation. The application introduces a structured Submitter–Approver workflow within a centralized platform, enabling simultaneous PowerPoint editing and controlled approval mechanisms. By digitizing and centralizing the entire process, CAPTURE significantly improves report accuracy, traceability, and operational efficiency.

Responsibilities:

- Played a major role in the end-to-end development of the CAPTURE application using Django, HTML, CSS, and JavaScript.
- Designed and implemented a structured Submitter–Approver workflow to replace manual email-based approval processes.
- Developed functionality to support simultaneous PowerPoint editing within the application, improving collaboration and version control.
- Built backend logic to manage report consolidation, approval tracking, and user access control.
- Eliminated dependency on fragmented email communication, centralizing the entire committee reporting lifecycle within a single platform.
- Improved efficiency, transparency, and governance in the committee report preparation process.
- Collaborated closely with business stakeholders to gather requirements and ensure seamless adoption of the application.

Project Title: Auto Collect Tool – Customer Data Collation for Risk Review

Description: The Auto Collect Tool was developed for the BUK team to streamline the process of preparing customer evaluation drafts for Risk Managers in Glasgow. Previously, stakeholders manually navigated through the FullServe and GPT portals to collect required customer data, consolidate it, and prepare draft documentation for transaction risk evaluation. This manual approach was time-consuming, repetitive, and operationally inefficient. The solution automated the end-to-end data collation process by integrating directly with both portals via APIs. Upon entering a Customer ID, the system retrieves relevant data from both sources and generates a consolidated PDF report within 30+ seconds, significantly improving turnaround time and operational efficiency.

Responsibilities:

- Designed and developed a fully API-based automation solution to extract and consolidate customer data from FullServe and GPT portals.
- Implemented logic to dynamically generate structured PDF reports based on stakeholder input (Customer ID).
- Reduced manual effort by eliminating the need to switch between multiple portals for data collection.
- Enabled report generation within approximately 30 seconds per customer.
- Delivered an automation solution currently used by 40+ team members on a daily basis.
- Delivered operational savings equivalent to 3 FTEs.
- Mentored and guided a campus graduate during implementation, ensuring successful completion and knowledge transfer.
- Collaborated with stakeholders and Risk teams to refine requirements and ensure alignment with business objectives.

Project Title: Project Eagle – Credit Risk Automation using Power Platform

Description: This project aimed to streamline and digitize the counterparty evaluation process used by credit risk teams. Previously performed entirely in Excel, the manual process involved collecting financial and reference data, building a credit model, and routing it for PRA (Pre-Risk Assessment) approval, which was error-prone, time-consuming, and lacked proper version control or auditability.

Responsibilities:

- Designed and implemented two Power Apps dashboards featuring Maker-Checker workflows, enabling efficient and controlled case processing.
- Built an Admin Dashboard to allow case assignment, approval tracking, and exporting of finalized credit models.
- Ensured full auditability by tracking user actions (who made changes, when, and where), fulfilling compliance and governance standards.
- Controlled data input through form validations and dropdown logic, reducing manual errors and improving data integrity.
- Replaced manual Excel workflows with a scalable, user-friendly app that reduced average processing time and increased traceability.
- Enabled structured case flow from Maker → Checker → Admin, reducing turnaround time and improving accountability across credit risk teams.

Project Title: CorpsAction – Time Series Adjustment for Asset Control
Description: The project addressed a data integrity issue in time series-based Profit and Loss calculations within the Asset Control system. When corporate actions like dividend issuance occurred, the resulting stock price drop was misinterpreted as actual movement, leading to incorrect PnL values. The goal was to ensure accurate adjustments by applying factors to reflect the true asset value changes.
Responsibilities:

- Analyzed existing issues in the Autosys job-based corporate action factor restructuring system and identified failure points impacting downstream teams.
- Rebuilt the entire workflow in Python, replacing legacy batch jobs with a more maintainable, modular, and scalable automation pipeline.
- Automated the application of adjustment factors to relevant time series data for each Asset Control ID, ensuring correct PnL calculation across scenarios.
- Developed logic to process and validate the adjustments dynamically, minimizing manual intervention and error rates.
- Implemented an automated email reporting system that sent cumulative status summaries and validation logs to stakeholders after each run.
- Enhanced reliability and maintainability of the solution, significantly reducing system failures and manual troubleshooting.

Project Title: STDF Analysis Automation – Corporate Banking
Description: This project aimed to improve the customer and facility-level analysis process within the Corporate Banking division. The existing solution, built on Teradata SQL, was limited to single-metric analysis, making it difficult for stakeholders to evaluate customer movement across multiple business dimensions.
Responsibilities:

- Designed and developed a Flask-based web application that dynamically analyses customer data across multiple dimensions (e.g., asset class, business class, customer maturity, etc.).
- Rewrote existing Teradata SQL logic in Python, enabling multidimensional filtering and analysis with improved performance and flexibility.
- Integrated the backend with Teradata databases to fetch real-time data and perform comprehensive customer journey tracking.
- Enabled tracking of customer movements between metrics such as matured vs. new clients, and asset class transitions.
- Delivered a fully dynamic, user-friendly UI to allow stakeholders to select analysis dimensions and download reports on demand.
- Successfully transformed a static, single-metric workflow into a scalable, business-oriented analytics tool supporting multiple KPIs.

Project Title: RISPI APP
Description: Rispi App is a web application designed to assist our client in generating statistical Excel reports for internal auditing purposes.
Responsibilities:

- Developed backend logic using the Django framework to handle data processing and generation of statistical reports.
- Implemented frontend components using HTML, CSS, and JavaScript to create an intuitive user interface for data input and report generation.
- Collaborated with the team to design and implement database models for efficient data storage and retrieval.
- Conducted testing and debugging to ensure the functionality and reliability of the application.
- Provided ongoing maintenance and support to address any issues or updates.

Project Title: BarX Error Report
Description: The BarX Error Report project aimed to streamline the identification and remediation of errors in asset control data, crucial for predicting risk volatility in the banking firm. Leveraging error reports from Outlook mail, we developed a Python-based solution to automate the process of identifying records requiring attention in asset control.
Responsibilities:

- Utilized Python libraries including Pywin32, OpenPyxl, and Pandas to scrape content from Outlook mail and extract relevant error data.
- Integrated with the Asset Control API to retrieve time series data and cross-reference it with error reports to identify missing dates and blank values.
- Implemented logic to generate detailed reports for risk analysts, highlighting errors and providing recommendations for remediation.
- Collaborated with stakeholders to refine requirements and ensure the solution met the needs of the risk analysis team.
- Conducted testing and debugging to validate the accuracy and reliability of the error identification and reporting process.

Project Title: Data Technique Weekly Check
Description: The Data Technique Weekly Check project aimed to ensure the integrity and compliance of six different data techniques used in the Asset Control application through regular checks and Email approval processes. This project was crucial for maintaining data accuracy and adhering to Barclays' audit policy.
Responsibilities:

- Developed a Python script utilizing libraries such as Pywin32, Pandas, OpenPyxl, PyODBC, and Beautiful Soup to automate the tracking of data operations and email approval processes.
- Implemented logic to track data updates in the Asset Control application and capture email sign-offs for data operations.
- Designed a comprehensive Excel report to summarize statistics including data update details, email approval status, and pending reviews.
- Collaborated with stakeholders to ensure compliance with Barclays' audit policy and incorporate necessary checks for data integrity.
- Conducted regular checks against a central file for audit purposes and recorded the results in the Excel report.

Project Title: European Banking Authority Q&A Scraper
Description: The European Banking Authority Q&A Scraper project aimed to automate the process of gathering regulatory Q&A data from the European Banking Authority website. This data was crucial for our client to stay updated on regulatory policies across Barclays. The project involved scraping approximately 1180 Q&A entries and downloading 40 associated files based on conditional checks.
Responsibilities:

- Developed Selenium scripts to navigate the European Banking Authority website and scrape regulatory Q&A data.
- Implemented conditional checks to download associated files based on specific criteria.
- Integrated data scraping and file downloading processes to automate the entire data gathering workflow.
- Designed logic to organize scraped Q&A data into an Excel file for quick reference by the client.
- Conducted testing and debugging to ensure the accuracy and reliability of the scraping and downloading processes.

Credit Suisse - Automation Engineer 2021 – 2023
Credit Suisse Group was a leading financial services company, advising clients in all aspects of finance, across the globe.

Project Title: Credit Suisse Resource Utilization Portal Scraper
Description: The Credit Suisse Resource Utilization Portal Scraper project aimed to automate the extraction of underutilized resource information from Credit Suisse's internal website. The project was initiated to support a program within our team focused on deleting unutilized resources in the bank's infrastructure. By scraping data from the resource utilization portal, we created a summary report to provide statistics for the program's progress.
Responsibilities:

- Developed Selenium scripts to navigate Credit Suisse's resource utilization portal and extract relevant information about underutilized resources.
- Implemented logic to filter and organize the extracted data into a summary report format.
- Successfully automated the extraction of underutilized resource information from Credit Suisse's internal portal, saving time and effort for the program.
- Generated a comprehensive summary report providing statistics on underutilized resources, enabling informed decision-making for resource deletion initiatives.

Project Title: AI Email Assistant
Description: The AI Email Assistant is an enterprise-level web application developed using React.js and Django, serving as a comprehensive solution for automating email categorization and processing. Its primary function is to onboard email inboxes and utilize an email classifier to categorize incoming emails based on predefined rules stored in MongoDB. The application encompasses the entire workflow, from mailbox onboarding to rule management and the execution of the email classifier script. Widely utilized across various teams within Credit Suisse, the AI Email Assistant significantly enhances email management efficiency and accuracy.
Responsibilities:

- Led the development of the frontend interface using React.js, ensuring an intuitive user experience for mailbox onboarding and rule management.
- Implemented backend functionalities using Django to handle user requests, interact with MongoDB for rule storage, and orchestrate the execution of the email classifier script.
- Collaborated closely with stakeholders from different teams to gather requirements and ensure alignment with business objectives.
- Conducted extensive testing and debugging to ensure the reliability and accuracy of email categorization and processing.

Project Title: Break Allocation
Description: Break Allocation is a Python-based process designed to automate the tagging of records in an Excel file according to predefined rules maintained in a central rule book. The process addresses the challenge of categorizing more than 15 categories of records, each with a substantial population that renders manual intervention impractical and time-consuming. By automating this task, Break Allocation significantly reduces manual effort, accelerates processing time, and ensures consistency in record categorization.
Responsibilities:

- Developed Python scripts to read data from Excel files and apply predefined rules for record tagging.
- Implemented logic to efficiently process large volumes of records across multiple categories.
- Collaborated with stakeholders to understand and implement business rules for record categorization.
- Conducted testing and validation to ensure accurate and reliable record tagging.

Project Title: KYC Screening Bot
Description: The KYC Screening Bot is an automation solution developed to streamline the manual screening process performed by KYC analysts for negative news screening of clients. Leveraging Selenium, pandas, and Natural Language Processing (NLP) techniques, the bot automatically checks the background of users applying for bank accounts at Credit Suisse. It analyzes client identities to identify any negative background information and generates a summary report for KYC analysts to further process.
Responsibilities:

- Developed Selenium scripts to automate the process of accessing client identity information from internal systems.
- Implemented NLP techniques to analyze news articles and other textual data for negative information related to clients.
- Collaborated with KYC analysts to understand screening requirements and refine the bot's functionality accordingly.

Project Title: Autonetics
Description: Autonetics is a web application developed using Angular and Java Spring Boot, designed to streamline the generation of Infrastructure as Code (IAC) templates and pipeline automation for Azure resources. The platform enables users to easily create pipelines for provisioning various resources in Azure, such as virtual machines, key vaults, and VMs with load balancers. Users can generate pipelines directly from the Autonetics interface and provision resources in Azure using either Jenkins or GitLab.
Responsibilities:

- Developed frontend components using Angular framework to create an intuitive user interface for pipeline generation and resource provisioning.
- Implemented backend functionalities using Java Spring Boot to handle user requests, generate IAC templates, and interact with Azure services.

Education

B. Tech IT (74%) — 2017 - 2021
Sri Venkateswara College of Engineering

H.S.C (87%) — 2016 - 2017
Christ Matric Higher Secondary School

S.S.L.C (95%) — 2014 - 2015
Christ Matric Higher Secondary School

Skills

Python Scripting	Troubleshooting & Debugging Code
Django	Linux
Selenium	Web Scraping
MongoDB & SQL	Java
HTML, CSS & JavaScript	Git
Devops	Multithreading & Multiprocessing
Restful APIs	Unit Testing (PyTest)
Power Automate	Microsoft Graph API
ReactJS	PowerApps